

Alex Liu

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Education

University of Florida (UF) - Gainesville, FL Expected Graduation: **May 2028**
Bachelor of Science in **Computer Science**, Minor in **Geography** **GPA: 3.75/4.0**
Relevant Coursework: Data Structures & Algorithms, Computer Org, Discrete Structures, Intro to Software Engineering

Skills

Languages: Python, C++, C, Java, JavaScript, TypeScript, Go, C#, SQL, RISC-V Assembly, Bash, HTML/CSS
Framework & Library: React.js, FastAPI, Flask, Node.js, OpenCV, MediaPipe, SQLAlchemy, NumPy, Pandas, TailwindCSS
Developer Tools: Git, GitHub, Docker, GitHub Actions (CI/CD), Linux/Unix, Postman, Vite, PyAutoGUI
Platforms & Databases: AWS (Lambda/ECS/S3/RDS), PostgreSQL, Vercel, Railway, Microservices, REST APIs, JWT

Projects

Geography Dashboard, React, AWS, Leaflet, FastAPI, Vite, TypeScript, PostgreSQL, Docker, Gemini AI **Nov 2025**

- Engineered a **full-stack** geospatial platform using a **microservices architecture** to enable **distributed deployment**, decoupling core GIS processing from **AI services** while ensuring **cross-cloud** observability by propagating X-Request-ID headers across FastAPI and AWS services.
- Architected a cloud-native AI engine on **AWS**, utilizing **HTTP APIs** to process structured spatial data (DBSCAN clusters, density grids) and generate interpretive summaries via Gemini API.
- Systematized a **CI/CD pipeline** via GitHub Actions to automate deployment, ensuring consistent configuration and **fault isolation** across cross-cloud environments while implementing database-backed **rate limiting**.
- Developed high-performance **REST APIs** with **FastAPI** and **SQLAlchemy** to process structured geospatial datasets(CSV/GeoJSON), transforming raw spatial results into interpretive analytical summaries for city-scale data.

UF Health SmartScribe, HTML, CSS, JavaScript, Supabase, Gemini AI **Oct 2025**

- Engineered a real-time **full-stack** medical documentation prototype within **24 hours**, integrating Deepgram and Gemini AI API to transcribe consultations and auto-generate structured SOAP notes.
- Architected a backend pipeline using **Supabase** to manage records, automating workflows to reduce time by **60%**.
- Implemented a speaker-aware AI extraction engine processing **60+ words/minute** with **sub-5-second** latency, prioritizing immediate physician utility over administrative overhead

SwampGuard — AI Insurance Claims Analysis Platform, React, FastAPI, Gemini, Python, Pydantic, RAG **Jan 2026**

- Designed an end-to-end claim intelligence workflow by integrating damage **photo analysis**, uploaded policy data, and itemized claim estimation logic, producing structured severity outputs and policy-adjusted repair cost projections
- Engineered **FastAPI** backend services by async image analysis endpoints and document extraction for receipts & insurance policies using JSON prompting & **Pydantic** validation, enabling consistent results across the **AI pipeline**
- Developed policy-aware claim calculation and legal guidance by **Xactimate-style** pricing, deductible and coverage logic tied to user's policies, and statute-grounded **RAG Q&A**, improving estimate interpretability and claim readines

Ability Bridge — Computer Vision Accessibility Interface, Python, OpenCV, MediaPipe, PyAutoGUI **Sep 2025**

- Engineered a hands-free accessibility interface utilizing **OpenCV**, processing **14,000+** facial landmarks/second to enable precise **real-time cursor control** and **Morse-code** text entry.
- Optimized signal processing via **exponential smoothing** and **hysteresis-based calibration** to achieve **30 FPS**, effectively eliminating cursor jitter and false triggers across diverse lighting conditions.

Experiences & Involvements

SASE Intern - UF Society of Asian Scientists & Engineers (SASE) **Aug 2025 - Dec 2025**

- Coordinated professional development events for **150+ attendees** by managing logistics and cross-functional communication, strengthening member career readiness and organizational programming.

Event Committee Member - UF Chinese American Student Association (CASA) **Aug 2025 - Oct 2025**

- Spearheaded** a large-scale cultural festival for **300+ participants**, overseeing program design and day-of execution.
- Curated** and managed cultural activities to increase student engagement and celebrate heritage for the UF community.

Instructor - Logic Lab **June 2022 – July 2022**

- Led and instructed a class of **30+** students in math and chess, designing lessons with **4** other team members.